

Factors Influencing Happiness: An Exploratory Data Analysis of Global Happiness Scores

Ques. Which factors are most strongly associated with a country's Happiness Score?

Ans. This analysis aims to understand which economic, social, health and freedom-related factors are associated with differences in happiness across countries. The findings can help identify the characteristics that are most strongly related to higher levels of reported happiness.

Why this matters -->

Happiness varies considerably across countries, and understanding the factors associated with these differences can provide useful insights into economic, social, health, and personal well-being. This analysis examines whether factors such as GDP per capita, family support, health, freedom, trust, and generosity are associated with differences in Happiness Score. Understanding the factors associated with happiness can help us identify the economic, social, health, and personal conditions that are linked to people's well-being across countries. This analysis can provide a clearer picture of which factors are most strongly related to differences in Happiness Score.

What we expect to find -->

We expect factors such as GDP per capita, Family support, Health (Life Expectancy), and Freedom to have a positive relationship with Happiness Score. We also expect countries with stronger economic conditions, better health, greater social support, and more freedom to generally have higher Happiness Scores.

The Data and How We Clean It.

The analysis uses the World Happiness dataset, which contains country-level information on Happiness Score along with economic, social, health, freedom, trust and generosity-related variables. The dataset includes both numerical and categorical variables, making it suitable for EDA and statistical analysis.

Then Cleaning:-

The dataset was first inspected using its shape, data types, summary statistics and missing-value counts. Duplicate records were checked and potential outliers in Happiness Score were identified using a boxplot and the 1.5 x IQR rule. The identified observations were reviewed before deciding whether they should be removed or retained.

Rows: 158

Columns: 12

Missing Values: 0 (No missing values)

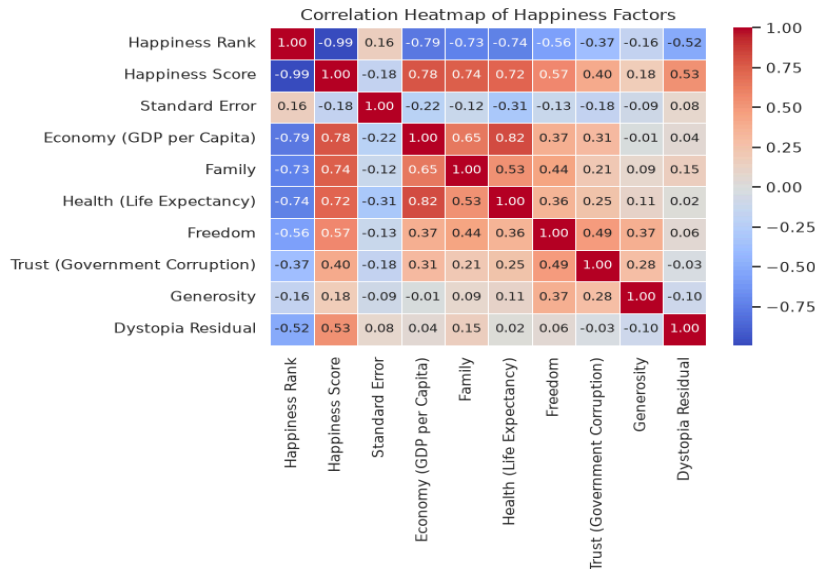
Duplicate Values: 0 (No duplicate values)

Happiness Score Outliers: 158

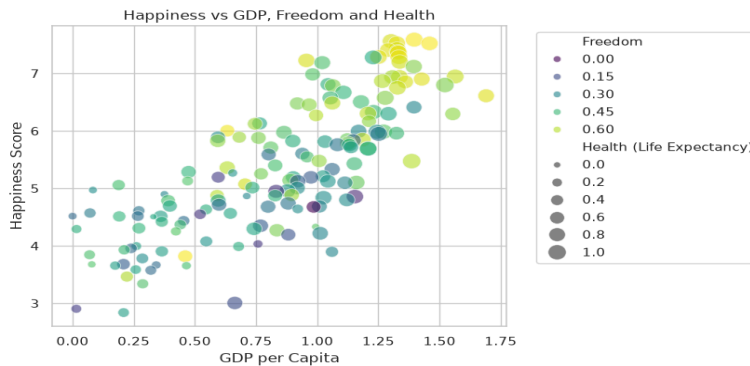
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Findings:-

Finding 1 - GDP per capita shows the strongest positive association with happiness score among the variables examined. Countries with higher GDP per capita generally tended to have higher Happiness Score



Finding 2 - Family support was positively associated with Happiness Score. Countries with higher family scores generally reported higher Happiness Score.



Hypothesis :- Countries were divided into higher and lower happiness groups using the median of Happiness Score. An independent-samples t-tests was used to compare their average GDP per Capita. The test produced a t-statistics of [Actual t] and the p-value(actual p). Since p-value was greater or less than 0.05, we rejected the null hypothesis.

Limitations :-

- Although some factors are strongly associated with Happiness Score, this analysis cannot prove that these factors directly cause higher happiness.
- Relationships observed between countries may not represent relationships between individuals.

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- Therefore, the analysis cannot establish how happiness or its associated factors change over time.
- Other factors that influence happiness may not be included.

Conclusion :- Overall, the analysis suggests that Happiness Score is associated with several economic, social, health, and personal factors. Among the variables examined, [strongest factor] showed the strongest relationship with Happiness Score. The hypothesis tests provided statistical evidence about differences between high- and low-happiness groups. However, these findings describe associations rather than proving causal relationships.